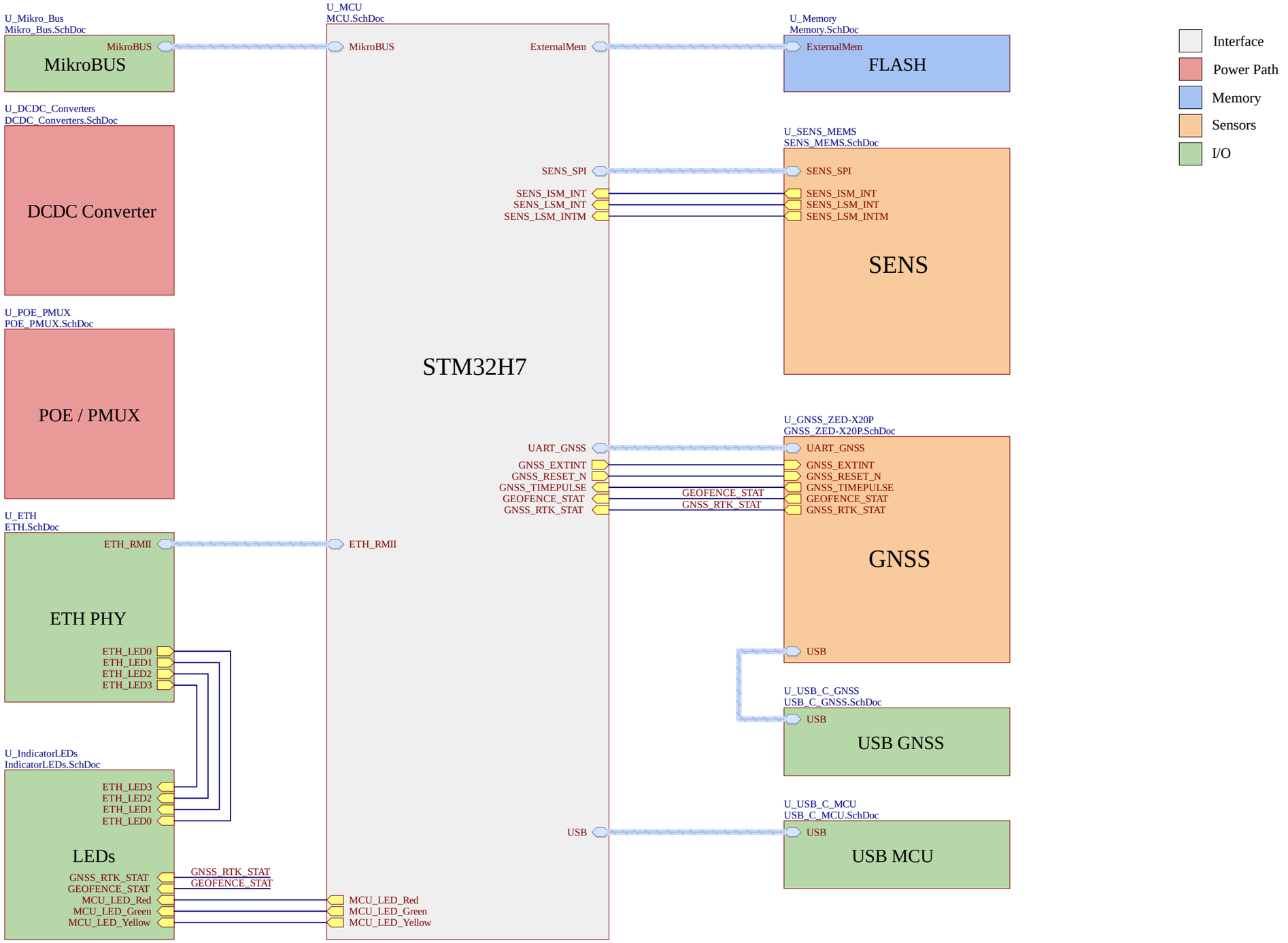
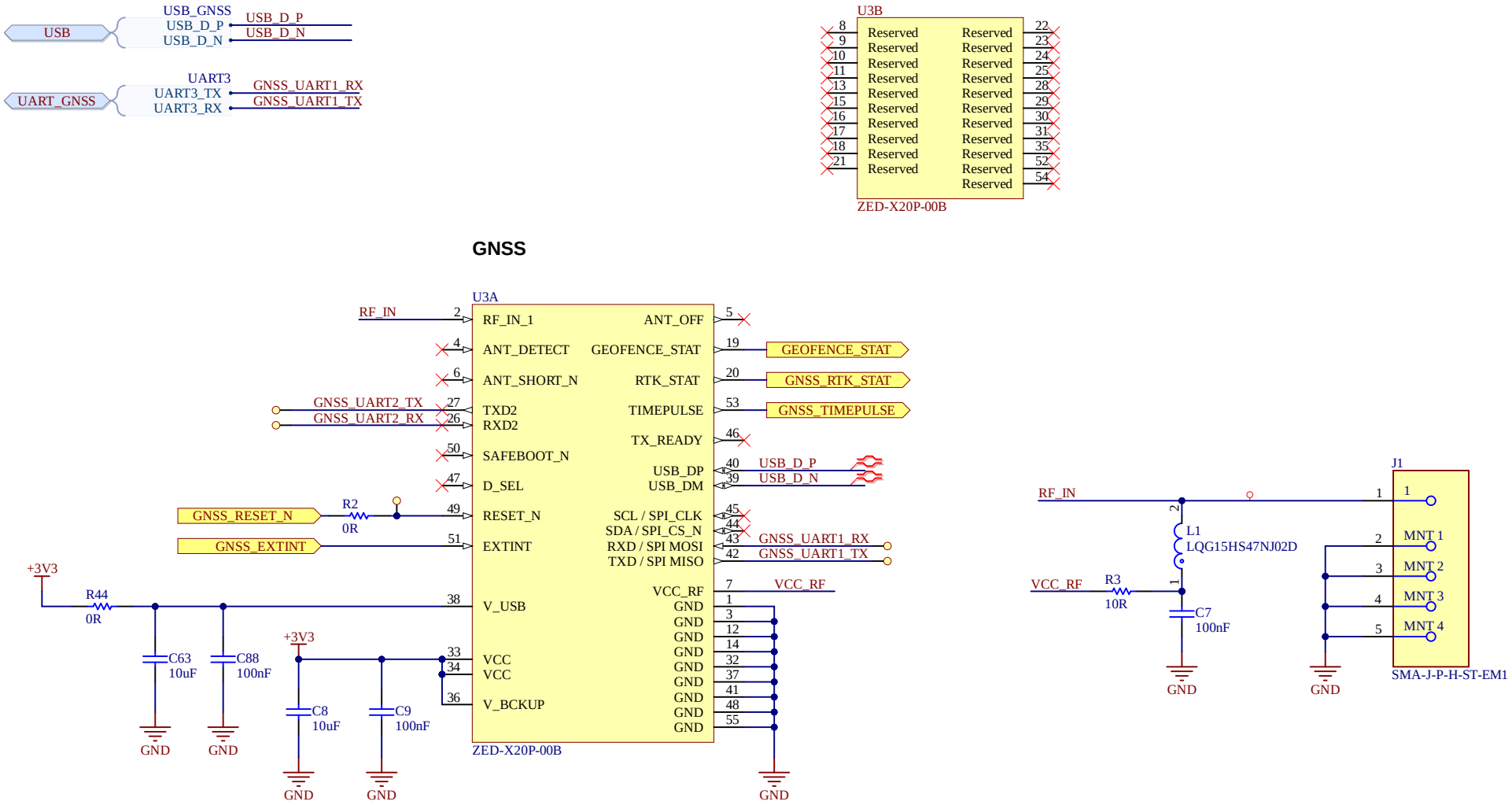


Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

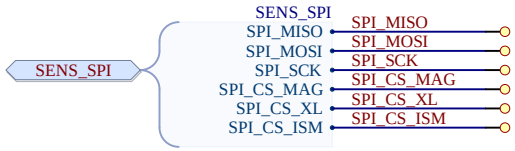
Project: Ublox GNSS Node

Drawing number: 1	Rev: A	Format: A3	Laboratory: PBL	Sheet: MAIN.SchDoc	
Date: 12/04/2026	13:32:47		Drawn by: mayerph/moosmannju/cclerc	Page: 1	of 12

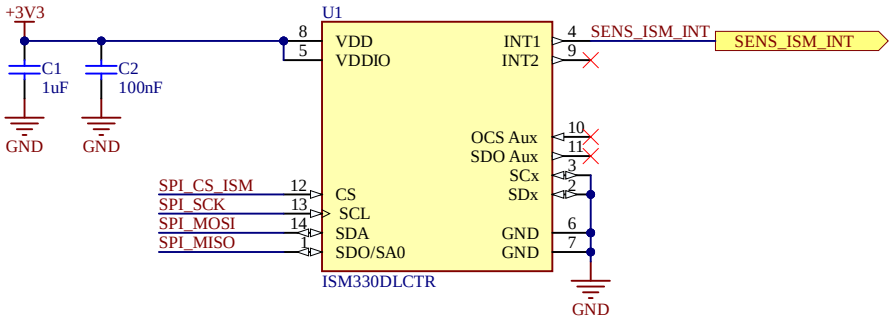
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\MAIN.SchDoc



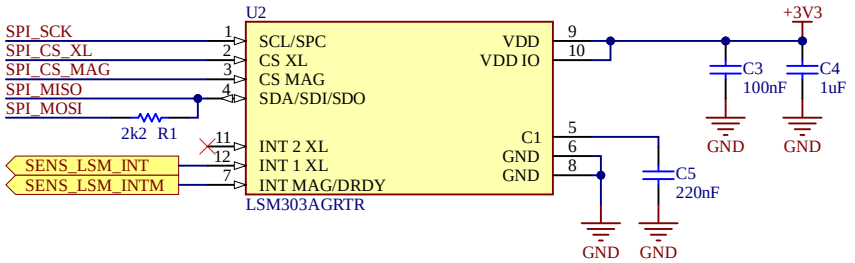
Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich	Project: Ublox GNSS Node		
Drawing number: 1 Rev: A Date: 27/06/2026 13:32:47	Format: A4	Laboratory: PBL Drawn by: mayerph/moos mannju/cclerc	Sheet: GNSS_ZED-X20P.SchDoc Page: 2 of 12
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\GNSS_ZED-X20P.SchDoc			



Gyro, Accelerometer

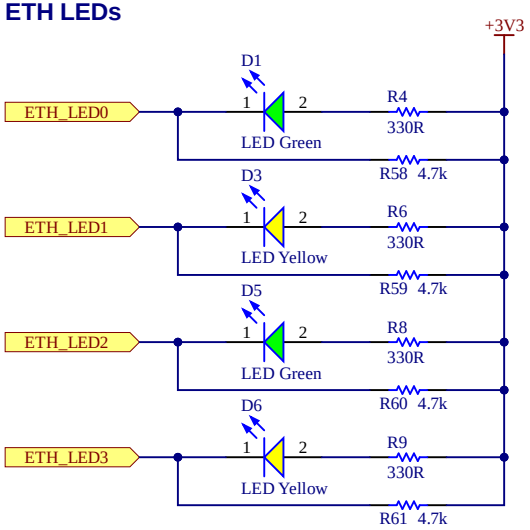


eCompas, Accelerometer Ultra-wide Bandwidth

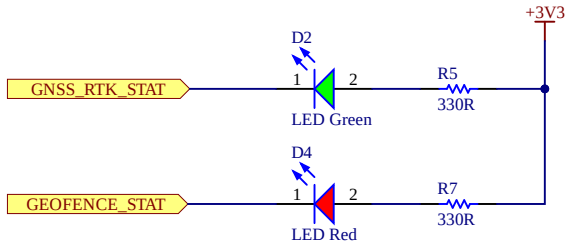


ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich	Project: Ublox GNSS Node			
Drawing number: 1 Rev: A Date: 27/06/2026 13:32:47	Format: A4	Laboratory: PBL Drawn by: mayerph/moos mannju/cclerc	Sheet: SENS_MEMS.SchDoc Page: 3 of 12	
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\SENS_MEMS.SchDoc				

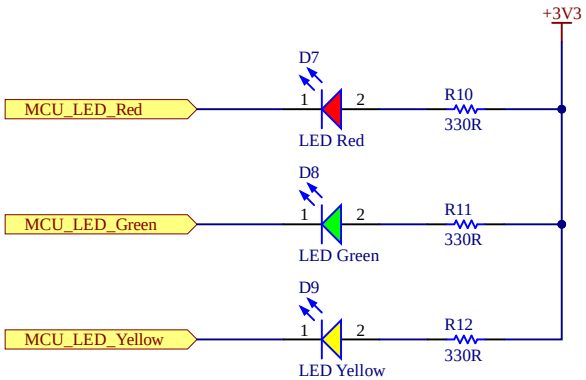
ETH LEDs



GNSS LED



MCU Status LEDs

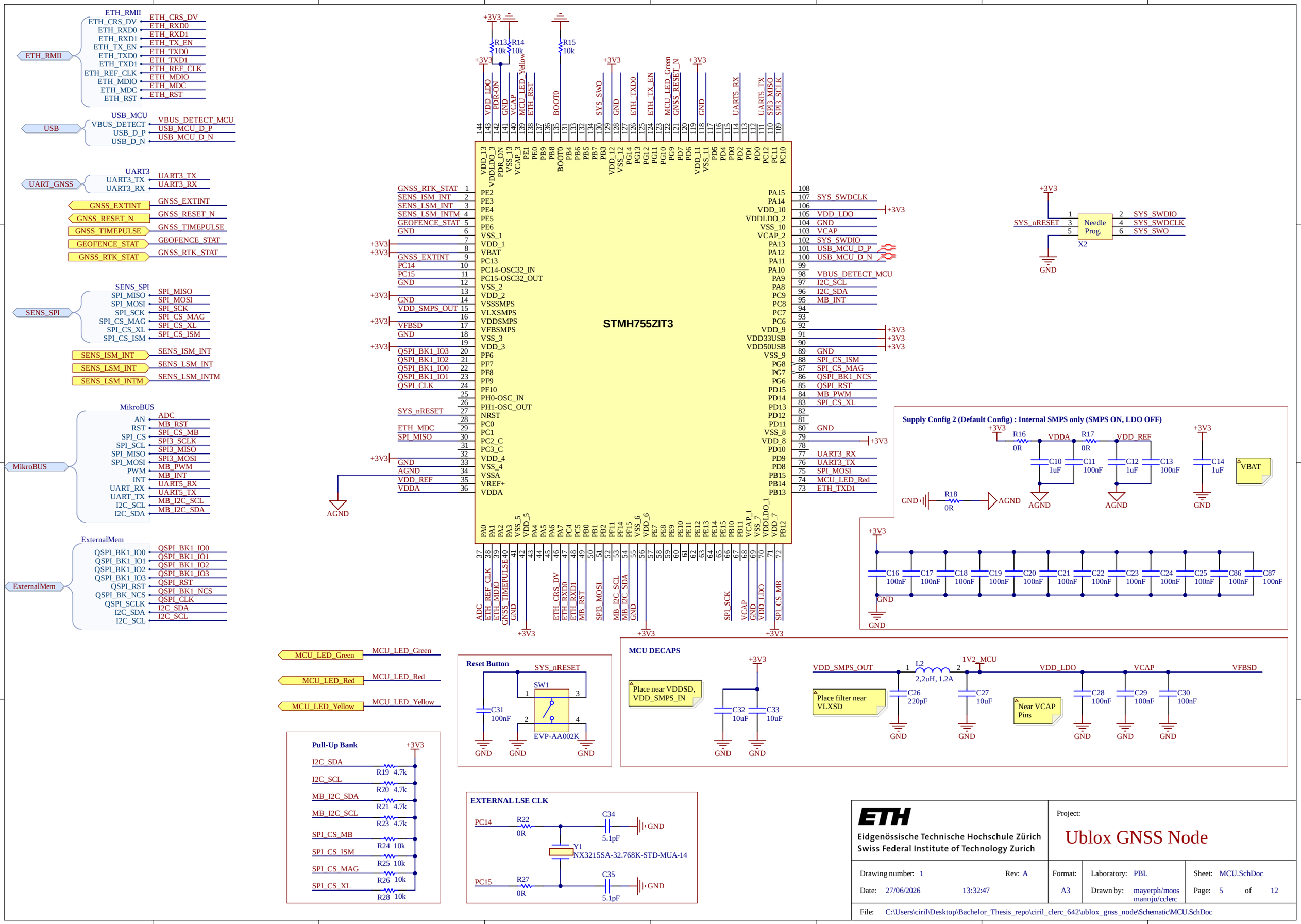


Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project: Ublox GNSS Node

Drawing number: 1	Rev: A	Format:	Laboratory: PBL	Sheet: IndicatorLEDs.SchDoc	
Date: 14/05/2026	13:32:47	A4	Drawn by: mayerph/moos mannju/cclerc	Page: 4	of 12

File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\IndicatorLEDs.SchDoc


ETH_RMII
ETH_CRS_DV
ETH_RXD0
ETH_RXD1
ETH_TX_EN
ETH_TXD0
ETH_TXD1
ETH_REF_CLK
ETH_MDIO
ETH_MDC
ETH_RST
USB
USB_MCU
VBUS_DETECT
USB_D_P
USB_D_N
VBUS_DETECT_MCU
USB_MCU_D_P
USB_MCU_D_N
UART_GNSS
UART3
UART3_TX
UART3_RX
GNSS_EXTINT
GNSS_RESET_N
GNSS_TIMEPULSE
GEOFENCE_STAT
GNSS_RTK_STAT
SENS_SPI
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_CS_MAG
SPI_CS_XL
SPI_CS_ISM
SENS_ISM_INT
SENS_LSM_INT
SENS_LSM_INTM
MikroBUS
AN
RST
SPI_CS
SPI_SCL
SPI_MISO
SPI_MOSI
PWM
INT
UART_RX
UART_TX
I2C_SCL
I2C_SDA
ADC
MB_RST
SPI_CS_MB
SPI3_SCLK
SPI3_MISO
SPI3_MOSI
MB_PWM
MB_INT
UART5_RX
UART5_TX
MB_I2C_SCL
MB_I2C_SDA
ExternalMem
QSPI_BK1_IO0
QSPI_BK1_IO1
QSPI_BK1_IO2
QSPI_BK1_IO3
QSPI_RST
QSPI_BK_NCS
QSPI_SCLK
QSPI_BK1_NCS
QSPI_CLK
R13 10k
R14 10k
R15 10k
+3V3
VDD_LDO
PDR-ON
GND
VCAP
MCU_LED_Yellow
ETH_RST
BOOT0
SYS_SWO
ETH_TXD0
ETH_TX_EN
MCU_LED_Green
GNSS_RESET_N
UART5_TX
SPI3_MISO
SPI3_SCLK
STMH755ZIT3
GNSS_RTK_STAT
SENS_ISM_INT
SENS_LSM_INT
SENS_LSM_INTM
SYS_nRESET
ETH_MDC
SPI_MISO
SPI_MOSI
SPI_CS_ISM
SYS_SWDCLK
VDD_10
VDDLDO_2
VSS_10
VCAP_2
SYS_SWDIO
USB_MCU_D_P
USB_MCU_D_N
VBUS_DETECT_MCU
I2C_SCL
I2C_SDA
MB_INT
VDD_9
VDD33USB
VDD50USB
VSS_9
SPI_CS_MAG
QSPI_BK1_NCS
QSPI_RST
MB_PWM
SPI_CS_XL
VSS_8
VDD_8
UART3_RX
UART3_TX
SPI_MOSI
MCU_LED_Red
ETH_TXD1
AGND
Needle Prog.
X2
SYS_SWDIO
SYS_SWDCLK
SYS_SWO
Supply Config 2 (Default Config) : Internal SMPS only (SMPS ON, LDO OFF)
R16 0R
R17 0R
R18 0R
VDDA
VDD_REF
C10 1uF
C11 100nF
C12 1uF
C13 100nF
C14 1uF
VBAT
C16 100nF
C17 100nF
C18 100nF
C19 100nF
C20 100nF
C21 100nF
C22 100nF
C23 100nF
C24 100nF
C25 100nF
C86 100nF
C87 100nF
MCU_LED_Green
MCU_LED_Red
MCU_LED_Yellow
Reset Button
SW1
C31 100nF
EVP-AA002K
MCU DECAPS
Place near VDDSD, VDD_SMPS_IN
C32 10uF
C33 10uF
VDD_SMPS_OUT
L2
2,2uH, 1.2A
1V2_MCU
VDD_LDO
VCAP
VFBSD
Place filter near VLXSD
C26 220pF
C27 10uF
Near VCAP Pins
C28 100nF
C29 100nF
C30 100nF
Pull-Up Bank
R19 4.7k
R20 4.7k
R21 4.7k
R23 4.7k
R24 10k
R25 10k
R26 10k
R28 10k
EXTERNAL LSE CLK
PC14
PC15
R22 0R
R27 0R
C34 5.1pF
C35 5.1pF
Y1
NX3215SA-32.768K-STD-MUA-14
ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich
Project:
Ublox GNSS Node
Drawing number: 1
Rev: A
Date: 27/06/2026
13:32:47
Format: A3
Laboratory: PBL
Drawn by: mayerph/moosmannju/cclerc
Sheet: MCU.SchDoc
Page: 5 of 12
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\MCU.SchDoc

ETH

ETH_RMII
ETH_CRS_DV
ETH_RXD0
ETH_RXD1
ETH_TX_EN
ETH_TXD0
ETH_TXD1
ETH_REF_CLK
ETH_MDIO
ETH_MDC
ETH_RST

PHY

U4
DP83826ERHBR
TX_CLK
TX_EN
TX_D0
TX_D1
TX_D2
TX_D3
RX_CLK/50MHz_RMII
RX_DV/CRS_DV
RX_D0
RX_D1
RX_D2
RX_D3
RX_ER
PWRDN/INT
MDIO
MDC
RST_N
VDDA3V3
VDDIO
TD_P
TD_M
RD_P
RD_M
LED0
CLKOUT/LED1
COL/LED2/GPIO
CRS/LED3
XI/50MHzIn
XO
RBIAS
CEXT
ModeSelect
GND
R29 2k2
R30 2k2
R31 2k2
R35 2k2
C53 10nF
C36 10nF
C37 100nF
C38 1uF
C39 10uF
C40 10nF
C41 100nF
C42 1uF
C43 10uF
FB1 BLM18KG601SN1D
FB2 BLM18KG601SN1D
+3V3
0R R32
R33 DNP
C50 100nF
R34 6k49
C48 22pF
C49 22pF
Y2 ABM8AIG-25.000MHZ-12-2Z-T3
U5 , make sure that for example pin 10 and one are connected (not through IC)
U5
TPD4E05U06QDQARQ1
D1+
D1-
D2+
D2-
NC
GND
PMD_TD_P
PMD_TD_N
PMD_RD_P
PMD_RD_N
ETH_LED0
ETH_LED1
ETH_LED2
ETH_LED3

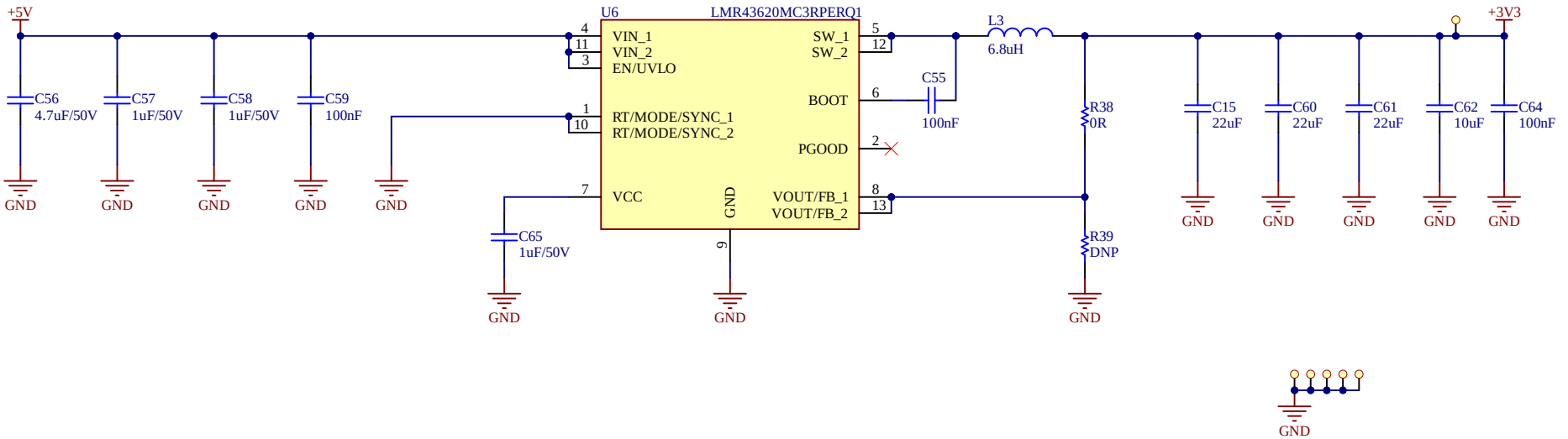
Galvanic Isolation

T1
SM91072AL
TD+
TD-
CT
TX+
TX-
RD+
RD-
RX+
RX-
C44 100nF
C45 DNP
C46 100nF
C47 DNP
X1
M12D-04PFFR-SF7005
TX_P
TX_N
RX_P
RX_N
POE_RX
POE_TX
C51 10nF/200V
C52 10nF/200V
R36 75R
R37 75R
GND_ETH
C54 1nF/2kV

ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich			Project: Ublox GNSS Node	
Drawing number: 1 Rev: A		Format:	Laboratory: PBL	Sheet: ETH.SchDoc
Date: 27/04/2026	13:32:47	A3	Drawn by: mayerph/moos mannju/cclerc	Page: 6 of 12
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\ETH.SchDoc				

DCDC

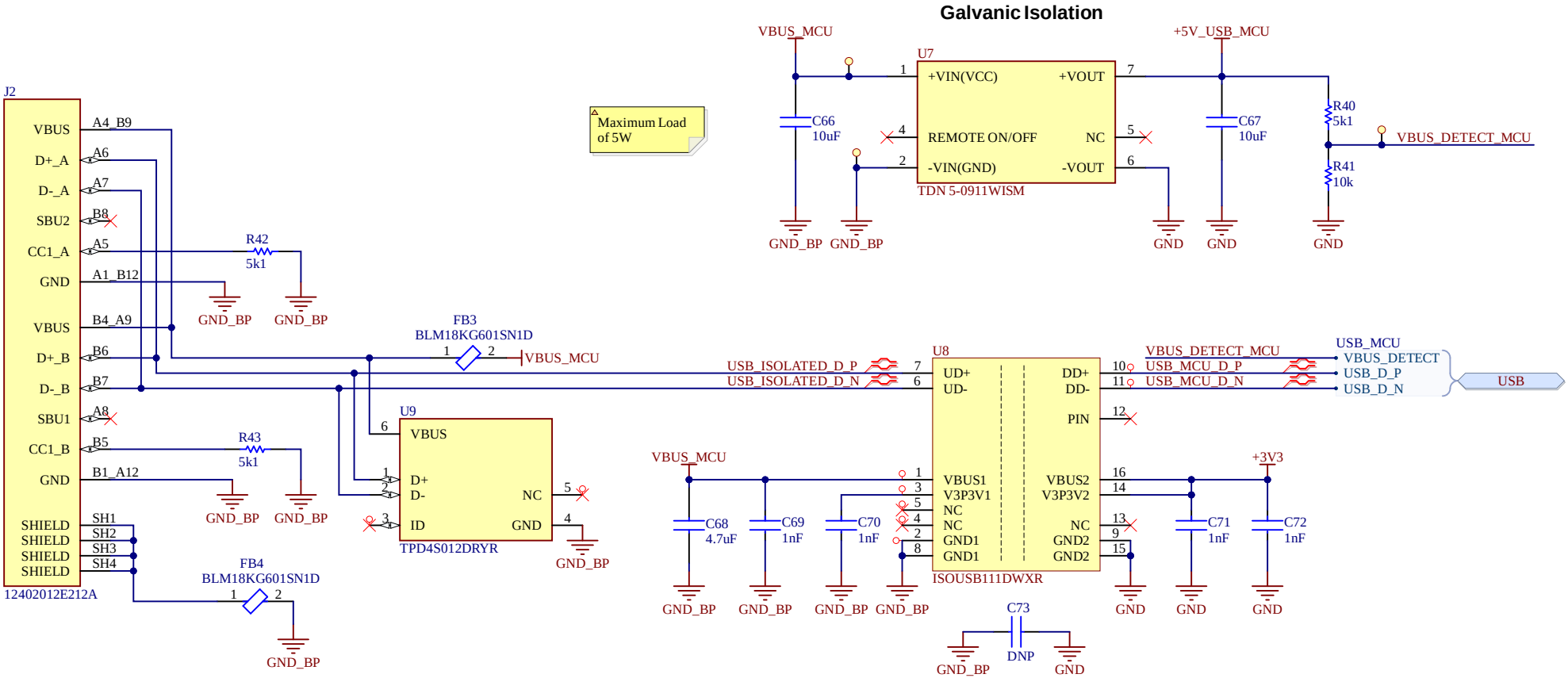
5V / 3.3V



ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich			Project: Ublox GNSS Node	
Drawing number: 1 Rev: A		Format:	Laboratory: PBL	Sheet: DCDC_Converters.SchDoc
Date: 23/04/2026 13:32:47		A4	Drawn by: mayerph/moosmannju/cclerc	Page: 7 of 12
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\DCDC_Converters.SchDoc				

USB type C for MCU

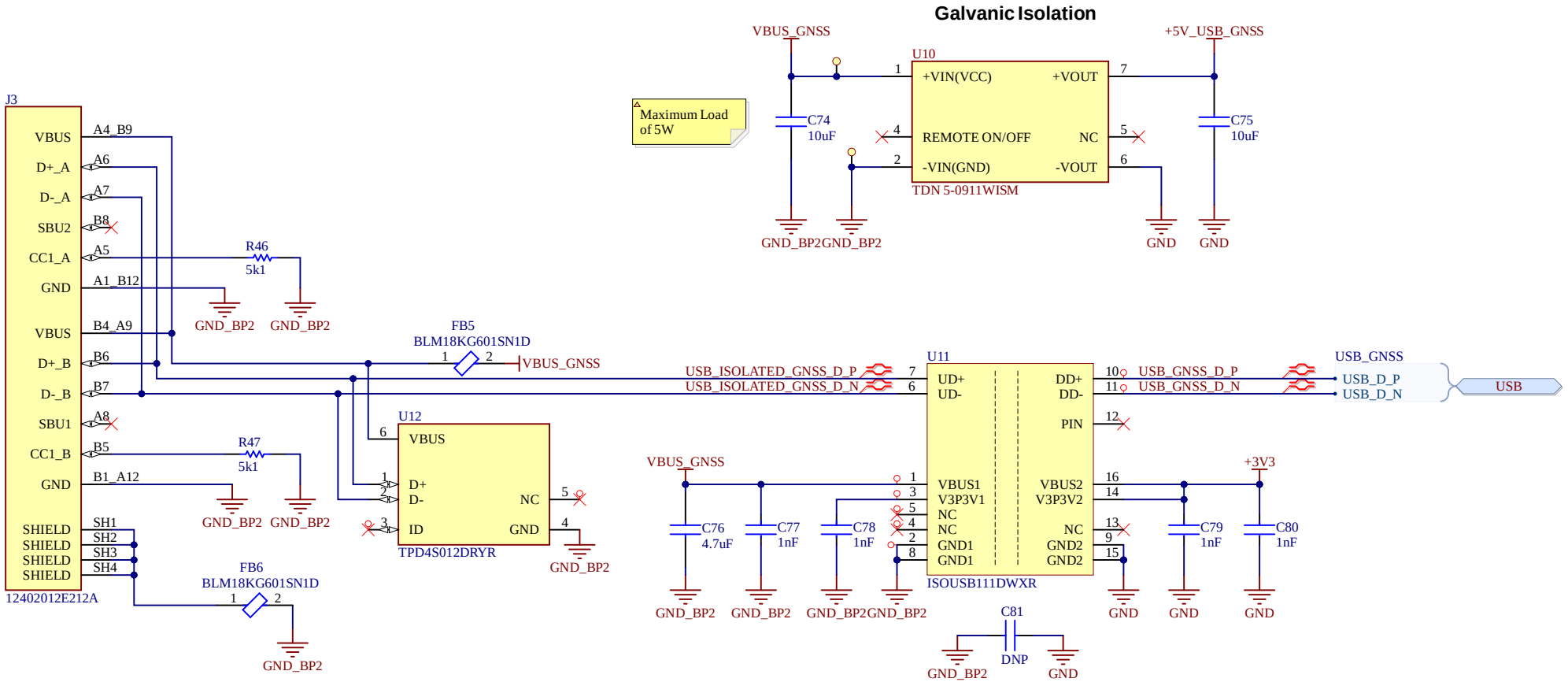
USB 2.0, +5V only



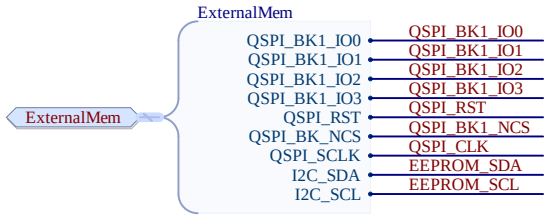
ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich			Project: Ublox GNSS Node	
Drawing number: 1 Rev: A		Format:	Laboratory: PBL	Sheet: USB_C_MCU.SchDoc
Date: 27/04/2026	13:32:47	A4	Drawn by: mayerph/moos mannju/cclerc	Page: 8 of 12
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USB type C for GNSS

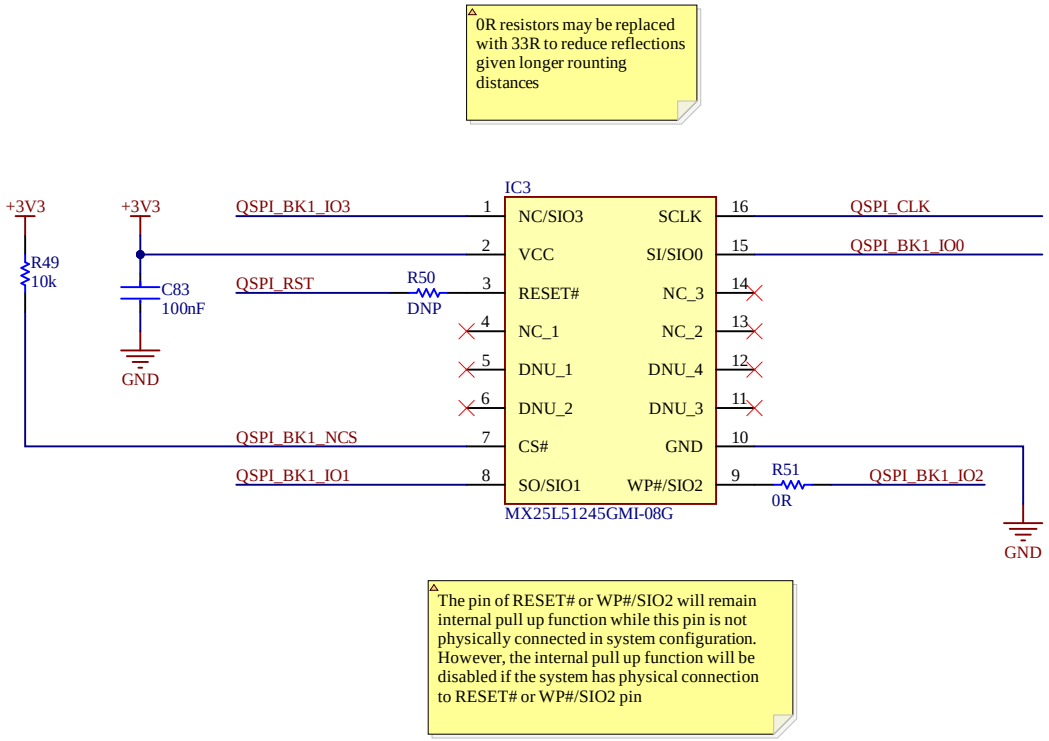
USB 2.0, +5V only



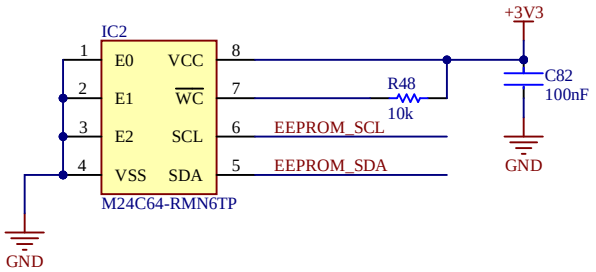
ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich			Project: Ublox GNSS Node	
Drawing number: 1 Rev: A		Format:	Laboratory: PBL	Sheet: USB_C_GNSS.SchDoc
Date: 14/05/2026 13:32:47		A4	Drawn by: mayerph/moos mannju/cclerc	Page: 9 of 12
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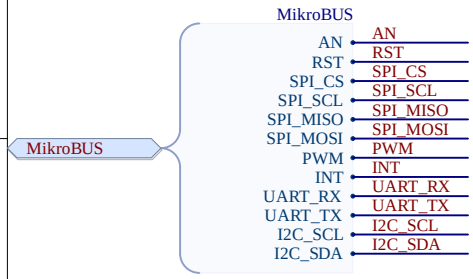
NOR FLASH QSPI

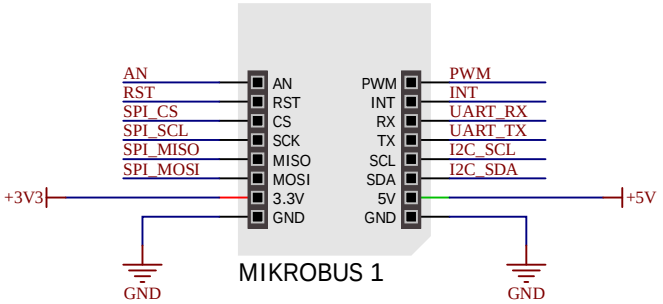


EEPROM



ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich			Project: Ublox GNSS Node	
Drawing number: 1 Rev: A		Format:	Laboratory: PBL	Sheet: Memory.SchDoc
Date: 27/06/2026	13:32:47	A4	Drawn by: mayerph/moos mannju/cclerc	Page: 10 of 12
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\Memory.SchDoc				



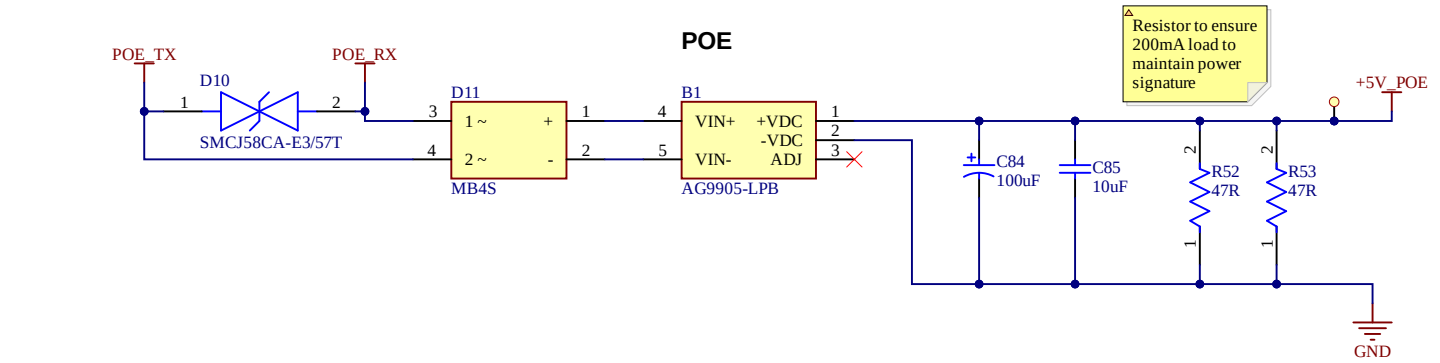


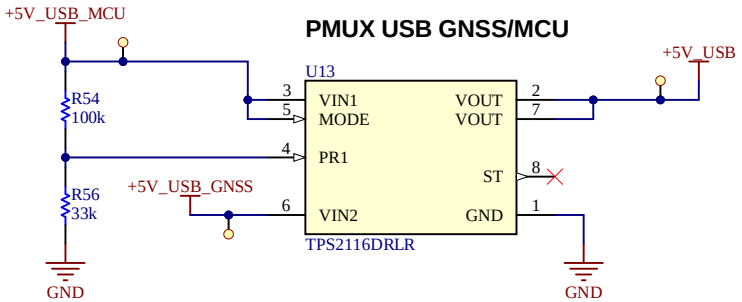
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

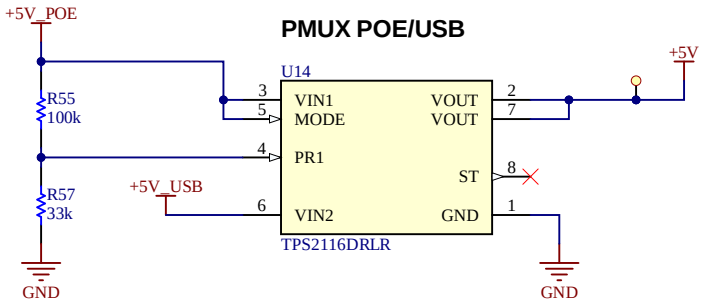
Project: Ublox GNSS Node

Drawing number: 1	Rev: A	Format: A4	Laboratory: PBL	Sheet: Mikro_Bus.SchDoc
Date: 14/05/2026 13:32:48			Drawn by: mayerph/moos mannju/cclerc	Page: 11 of 12

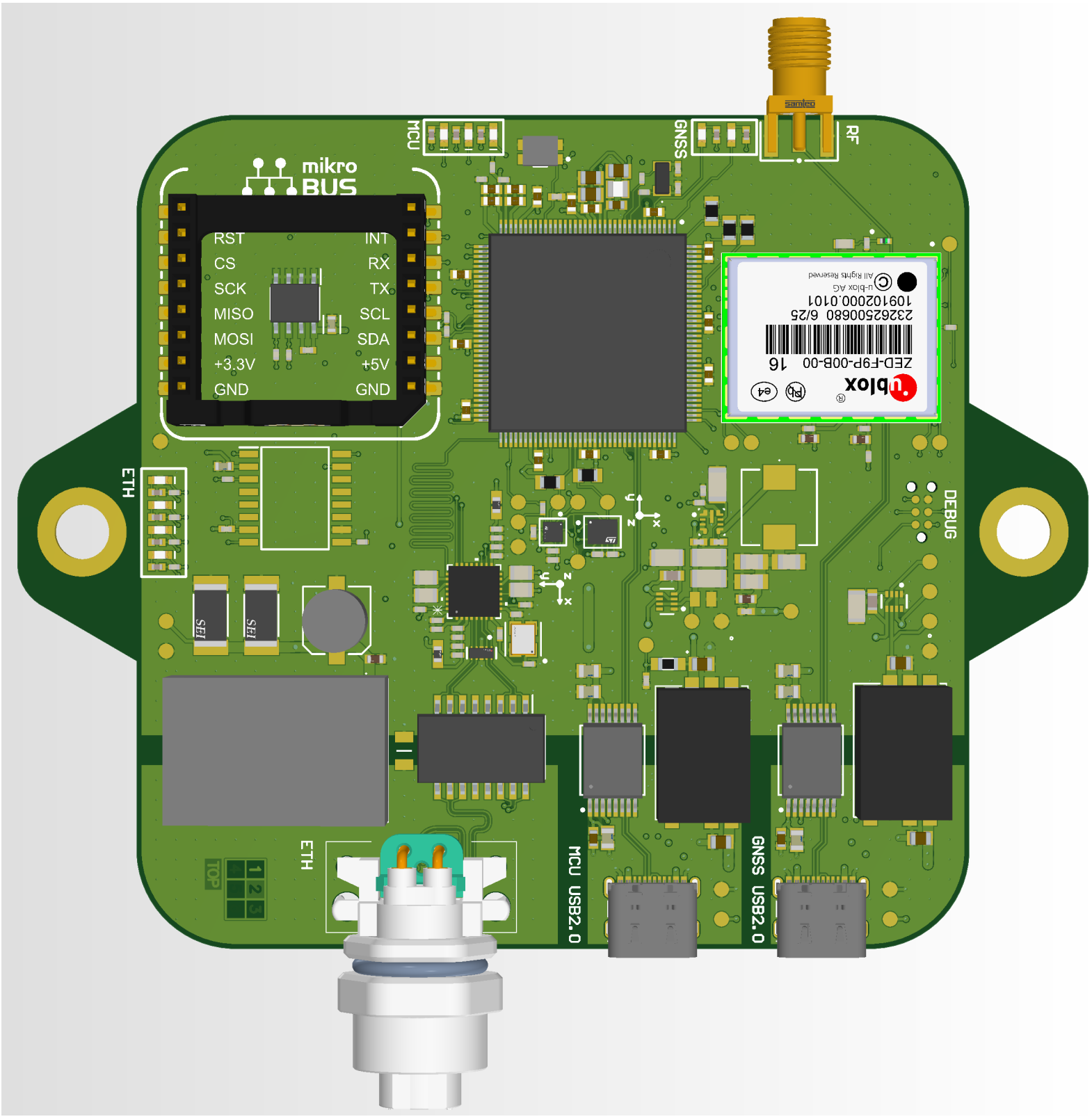
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Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich			Project: Ublox GNSS Node		
Drawing number: 1	Rev: A	Format: A4	Laboratory: PBL	Sheet: POE_PMUX.SchDoc	
Date: 27/06/2026 13:32:48			Drawn by: mayerph/moos mannju/cclerc	Page: 12 of 12	
File: C:\Users\ciril\Desktop\Bachelor_Thesis_repo\ciril_clerc_642\ublox_gnss_node\Schematic\POE_PMUX.SchDoc					


MCU
GNSS
RF
mikro BUS
RST
CS
SCK
MISO
MOSI
+3.3V
GND
INT
RX
TX
SCL
SDA
+5V
GND
ZED-F9P-00B-00
u-blox AG
All Rights Reserved
ETH
DEBUG
SEI
SEI
ETH
TOP
MCU USB2.0
GNSS USB2.0

Comment	Description	Designator	Footprint	LibRef	Quantity	Manufacturer	Manufacturer Part Number	Part Number	PartNumber	Manufacturer Name	Manufacturer_Name	Supplier	Supplier Part Number
2ED-X03F-008	p-block X03 all-band GNSS High Precision module LGA, 250 pcs/reel,	U3	2ED-FIP	2ED-X03F-008	1	u-blox	2ED-X03F-008	2ED-FIP-008				Digikey	672-2ED-X03F-008CT-ND
TPS2116DRJR	1.6-V to 5.5-V, 40-mA,	U13, U14	DRJ0008A-MFG	TPS2116DRJR	2		TPS2116DRJR			Texas Instruments			
TFD45EL2JE80	TVS DIODE 5.5 VOLTA	U9, U12	FP-DRV0005A-MFG	CMP-0101-00038-4	2		TFD45EL2JE80						
TFD45EL60J(QQ)H(Q)1	TVS DIODE 5.5 V 14 V 45 UNION	U6	FP-DQN0010A-MFG	CMP-04585-000004-1	1	Texas Instruments	TFD45EL60J(QQ)H(Q),						
STM32H755ZIT6U	STM32H7X5 MCU	IC1	QFP50P2200X2200X2 300-144N	STM32H755ZIT6U	1		STM32H755ZIT6U			STMicroelectronics			
SMCJ58CA-E3/57T	TVS DIODE 58 V 93.6 V DO214AB	D10	FP-SMC-MFG	CMP-02577-000016-1	1	Vishay Semiconductor Diodes Division	SMCJ58CA-E3/57T						
SM6S1072AL	Telecom Transformer 1:1:1, 4 Ohm Prim. DCR 1 Ohm Sec. DCR 1 Terminal Gull Wing SMD,Automotive	T5	SOIC127P650X590-16N	SM6S1072AL	1	Bourns	SM6S1072AL					Digikey	118-SM6S1072AL-ECT-ND
47R	RTO xERL1WV2512 Thick Film Chip Resistor AEC-Q200 compliant	R52, R53	ST-RMCP2512-MFG	CMP-28527-000410-1	2	Stackpole Electronics	RMCP2512JT47R					Arrow	RMCP2512JT47R
NX3215SA-32.768K-STD-MUA-14	Crystal or Oscillator - SoT Load	Y5	NX3215SA2768K000-MUA6	NX3215SA-32.768K-STD-MUA-14	1		NX3215SA-32.768K-STD-MUA-14						
MX25L51245GM-08G	NOR Flash 512Mbit QSPI Macronix	IC3	SOIC127P810X305-16N	MX25L51245GM-08G	1		MX25L51245GM-08G					Macronix	
MB4S	SINGLE RECT 2 P40V 500MA MSOIC	D11	ST-TS12P-MFG	CMP-07143-000033-1	1	ON Semiconductor	MB4S						
KO4CS4-PdNiETP	ESTRON ST3 64bit OC	IC2	SOIC127P650X175-8N	KO4CS4-PdNiETP	1		KO4CS4-PdNiETP			STMicroelectronics			
MU2D-04RFRR-SF7003	Conn Circular F 4 POS Solder PA Panel Mount 4 Terminal 1 Port	X5	MU2D04RFRRSF7003	MU2D-04RFRR-SF7003	1	Amphenol	MU2D-04RFRR-SF7003						
LSM303AGRTR	IMU ACCELNMG GYCSPI 12SGA	U8	FP-LGA-12-848735-MFG	CMP-2000-07466-2	1	STMicroelectronics	LSM303AGRTR						
2.2uH, 1.2A	Inductor	L2	LQM2MPN2R2MG0L	LQM2MPN2R2MG0L	1		LQM2MPN2R2MG0L			Murata Electronics			
LQG15HS47NJ02D	Multilayer type RF inductor 47nH ±5% 0.72Ω 300mA 0402	L1	FP-LQG15HH_02-PC_B	CMP-06042-0157813-1	1	Murata	LQG15HS47NJ02D					Digikey	490-2631-6-ND
LM5163DXMC3RRE/NOPB	Integrated Circuit	U6	LM5163DXMC3RRE/NOPB	LM5163DXMC3RRE/NOPB	1		LM5163DXMC3RRE/NOPB			Texas Instruments			
10uF	Capacitor 10uF 0805	C37, C52, C53, C55, C57, C74, C75	CAPC2012N-L45N	LMK212ABT106MG-T	7		LMK212ABT106MG-T			TAIYO YUDEN			
ISOUSB111DWAR	Low-emissions, full- and low-speed isolated USB repeater (4) SOIC 40 to 125	U8, U11	DWA0016A	ISOUSB111DWAR	2	Texas Instruments	ISOUSB111DWAR					Digikey	296-ISOUSB111DWARCT-ND
GM3303CHCXR	RMMCHSERIAL MODULE, ALWAYS ON	X1	FP-LGA-14L-EM0024845-MFG	CMP-12177-000001-1	1	STMicroelectronics	GM3303CHCXR					Digikey	497-19784-6-ND
10uF	CAPC3261M1F 25V -55/+0833	C9, C62, C63, C65	FP-GRM188R61E106MA73	GRM188R61E106MA73	4	Murata	GRM188R61E106MA73D					Digi-Key	490-7202-6-ND
100nF	Capacitor 100nF 0402 - storage	C31, C59, C58, C64	CAPC1005X05-025N-LD	GRM155R71H104KE14	4	Murata	GRM155R71H104KE14D						
DNP	Capacitor 100nF 0402 - storage	C45, C47	CAPC1005X05-025N-LD	GRM155R71H104KE14	2	Murata	GRM155R71H104KE14D						
10nF	Capacitor 10nF/50V 0402 - storage	C34, C40, C53	GRM15x	GRM155R71H103KA88	3		GRM155R71H103KA88D			Murata Electronics			
100nF	[Yehalua] _Capacitor 100nF/0402 - storage	C7, C3, C7, C9, C37, C41, C44, C45, C50, C69	CAPC1005X05-025LNB-STD	CMP-2009-03928-1, GRM155R71H104KE14	10	Murata	GRM155R71C104KA88J, GRM155R71H104KE14,	GRM155R71C104KA88J [NoParnam]					
10uF	Capacitor 10uF/0402 - storage	C34, C43	GRM15x	GRM155R61A106ME15	2		GRM155R61A106ME15D			Murata Electronics			
1uF	Capacitor 1uF/35V 0402 - storage	C36, C42	CAPC1005X70N	GRM155R61V105KE15	2		GRM155R61V105KE15D			Murata Electronics			
1uF	CAP 1uF 0402 [1005 Metric]	C1, C4	CAPC0402(1005)0.L	CMP-1034-02114-1	2	Murata	GRM155C31C105KE01						
4.7uF/25V	Capacitor 4.7uF/25V 0805	C56	CAPC2012X145N	GRM21BR61H475ME15	1		GRM21BR61H475ME15L			Murata Electronics			
EVP-AK02K	Light Touch Switch, 200 mOhm, -40 to 85 degC, 4 Pin SMD, RoHS, Tape and Reel	SW5	PNSC-EVPA_V	CMP-2000-06231-1	1	Panasonic	EVP-AK02K					Panasonic Industrial Devices	P13473DKRHD
5k1	Resistor 5k1 0805	R40, R42, R43, R46, R61	ERJ-P6	ERJ-UP6D5101V	5		ERJ-UP6D5101V			Panasonic			
10k	Resistor 10k 0805	R13, R14, R23, R24, R25, R26, R28	RESC2012X70N	ERJ-6GEYJ103V	7		ERJ-6GEYJ103V			Panasonic			
10k	Resistor 10k 0402	R41, R48, R49	RESC1005X40-025-LD-STD5	CMP-2002-00410-1	3	Panasonic	ERJ-3GEJ103X						
75R	Resistor 75R 0805	R56, R57	RESC2013X60-140N-LD-STD5	CMP-2001-00660-1	2		ERA4EB750V	ERA4EB750V					
2k2	Resistor 2k2 0603	R29, R30, R31, R35	RESC1608X55-060-LD-STD5	CMP-2001-00613-1	4	Panasonic	ERA4EB222V						
0.8uH	Inductor 0.8uH 2.55A 43.5 MOHM	L3	CRT3918R	CRT3-6R8-R	1		CRT3-6R8-R			Eaton			
DP83826ERHBR	Ethernet Transceiver, 3 Timer, RQC32	U4	QFN_ERHBR_TEX	DP83826ERHBR	1	Texas Instruments	DP83826ERHBR					Digikey	296-DP83826ERHBRCT-ND
0R	Resistor 0R0402	R2, R56, R17, R18, R31, R37, R38, R44, R51	RESC1005X40-025-LD-STD5	CMP-2002-03041-1	9	Vishay Dale	CRCW40200000Z0EDHP	CRCW40200000Z0EDHP					
0R	Resistor 0R0402	R32	RESC1005X40-025-LD-STD5	CMP-009-00163-1	1	Vishay Dale	CRCW40200000Z0EDHP						
DNP	Resistor 0R0402	R33	RESC1005X40-025-LD-STD5	CMP-009-00163-1	1	Vishay Dale	CRCW40200000Z0EDHP						
DNP	Resistor 0R0402	R39, R50	RESC1005X40-025-LD-STD5	CMP-2002-03041-1	2	Vishay Dale	CRCW40200000Z0EDHP	CRCW40200000Z0EDHP					
100k	100K 0.1W 1% 0603 (1608 Metric) SMD	R54, R55	RESC0603(1608)_N	CMP-1012-00735-1	2	Vishay, [NoParnam]	CRCW0603100KFKTC, [NoParnam]					Digikey, [NoParnam]	CRCW0603100KFKTC-ND, [NoParnam]
0k49	Resistor 0k49 0402 1%	R34	RESC1005X40-025-LD-STD5	CMP-2002-00186-1	1	Vishay Dale	CRCW40200K49FKED						
10R	Chip Resistor, 10 Ohm, +/- 1%, 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R9	RESC1608X55-025-N-LD-STD5	CMP-1659-00008-4	1	Yageo	CR0603-FX-10R0ELF					Digikey	CR0603-FX-10R0ELFCT-ND
1nF/2kV	Capacitor 1nF/2kV 3216 Isolated GND Loop Cap	C54	CAPC3216X160N	CL31B102KJHNNNE	1		CL31B102KJHNNNE			SAMSUNG			
DNP	Capacitor 1nF/2kV 3216 Isolated GND Loop Cap	C73, C81	CAPC3216X160N	CL31B102KJHNNNE	2		CL31B102KJHNNNE			SAMSUNG			
4.7uF	Capacitor 4.7uF/0603	C69, C76	CAPC1608X90N	CL10B475KQ8NQNC	2		CL10B475KQ8NQNC			SAMSUNG			
1uF	Capacitor 1uF/0603	C10, C12, C14	CAPC1608X90N	CL10B105KP8NNNC	3		CL10B105KP8NNNC			SAMSUNG			
220nF		C8	CAPC1005X55-025-LD-STD5	CMP-1034-02087-2	1	Samsung	CL05B224KO5NNNC					Digikey	1276-1176-1-ND
100nF	Capacitor 100nF/0603	C11, C13, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C29, C30, C66, C67	CAPC1608X90N	CC0603KRX7R7BB104	17		CC0603KRX7R7BB104			YAGEO			
1uF/50V	Capacitor 1uF/50V 0603	C27, C68, C85	CAPC3216X160N50-STD5	CMP-2000-06311-1	3	Kemet	C1206C105K5RACTU						
1nF	Capacitor 1nF/0805	C69, C70, C71, C72, C77, C78, C79, C80	TP-C0805C-DN-MFG	CMP-2007-00235-2	8	KEMET	C0805C102J5GACTU						
100nF	Capacitor 100nF/0603	C60, C83	TP-C0603C-CF-MFG	CMP-2006-03061-2	2	KEMET	C0603C104K5RACTU						
10nF/200V	CAP 10nF 200V ±10% 0603 [1608 Metric] Thickness 1mm SMD	C51, C52	CAPC0603(1608)0.N	CMP-1035-02328-1	2	Yageo Group	C0603C103K2RAC						
22pF	Capacitor 22pF/0402	C48, C49	C0402C104J4RACTU	C0402C220K5GACTU	2		C0402C220K5GACTU			KEMET			
BLM18PG601SN1D	Chip Ferrite Bead, 0603, 600Ω @ 100MHz, 0.15Ω, 25%, 3.2A	FB1, FB2, FB3, FB4, FB5, FB6	FP-BLM18-0_15-0_8-PC_A	CMP-0696-00581-3	6		BLM18PG601SN1D, [NoParnam]						
AQM905-MTB	Power over Ethernet - PoE PD Module, Isolated DC-DC converter, IEEE802.3at compliant, Over Temp. Protection, Ind Temp. Range, Small Footprint, SMT, (Blocks not pins)	B3	AQM905MTB	AQM905-MTB	1		AQM905-MTB			Bivertel			
ABMMAG-25.000MHZ-12-22-T3	Crystal 25MHz ±50ppm (Stability) 12pF FUND 60Ω Automotive 4-Pin Ultra Mini-C SMD T/R	Y2	FP-ABMMAG-MFG	CMP-27762-000325-2	1		ABMMAG-25.000MHZ-12-22-T3						
1240201352121A	2882.0, Type C, Top mount, CH 1.65mm, Dual Row SMT	J2, J3	AMPHENOL_1240201352121A	1240201352121A	2		1240201352121A						
220pF	Capacitor 220pF/0603	C26	CAPC1608X87N	0603B221K500CT	1		0603B221K500CT			Walsin Technology Corporation			
2k2		R1	RESC1005X40-025-LD-STD5	CMP-2002-07567-1	1	Yageo / Phycomp		RC0402FR-072K2L					
4.7k		R10, R20, R21, R22, R58, R59, R60, R61	RESC1005X40-025-LD-STD5	CMP-1011-00060-2	8	Vishay Dale		CRCW40274K70JNED					
1.1pF		C34, C35	CAPC1005X05-025-N-LD-STD5	CMP-2008-04302-1	2	Murata		GRM0555C1H1R1CA01D					
22uF		C15, C60, C61	CAPC3013X157-025-N-LD-STD5	CMP-2007-04522-1	3	Murata Electronics / Elco		490GZD226M4VA					
10k		R64, R67	RESC1005X40-025-LD-STD5	CMP-2003-03510-1	2	Yageo		RC1210KJR-1010KL					
100uF	Aluminum Electrolytic Capacitor, UD Series, Low Impedance, 100 uF, 25 V, +/- 20%, -55 to 105 degC - Chip Type, 6.3 x 7.7 mm Dia. - Tin-Free, Reel	C64	NIC-CAPAE-6.3X7.7_V	CMP-0687-00063-1	1	Nichicon		UUD1E101MCL1GS					
330R	330RΩ 063W/5%/0402 [1005 Metric] SMD	R8, R5, R6, R7, R9, R9, R10, R11, R12	RESC0402(1005)_M	CMP-1011-00062-1	9							Digikey	CRCW0402330RFKED
LED Green	MUR1HS1UR7RKNK-1500/0/VS7/000 - LED Green, SMD, 0603, 30 mA, 2 V, 570 nm	D3, D2, D5, D6	WL-SMCW_0603_150060x	CMP-1426-00006-7	4							RS	8154215P
LED Red	LED Low Power Ultra-Color Red 630nm 2V/2 mA Chip 0603(1608Metric) TR	D4, D7	WL-SMCW_0603_150060x	CMP-1426-00001-4	2							Farnell	1500609575000
LED Yellow	WL-SMCW SMT Micro-Color Chip LED Waterclear	D3, D6, D9	WL-SMCW_0603_150060x	CMP-1426-00005-3	3							Farnell	1500609575000
MI40KBLSHOST	microSLSHost Socket	MI40KBLS1	MI40KBLSHOST	MI40KBLSHOST	1							Digikey	1471-MI40KBLSHOSTCT-ND
SMA-JP-H-ST-EM1	CONN SMA JACK STR 50OHM EDGE MNT	J1	FP-SMA-JP-H-ST-EM1-MFG	CMP-07754-000006-1	1							Amet	SMA-JP-H-ST-EM1
TDN5-0911WSM	TDN 5-0911WSM DC-DC Converter, 5V, 1A, No. of Outputs: 1, Output, Output Power Max: 5W, Voltage - 1st Current 1: 1A	U7, U10	TDN1W5M SINGLE_TEP	TDN5-0911WSM	2							Digikey	TDN5-0911WSM-ND
testpoint		*7, *8, *10, *11, *13, *14, *15, *16, *17, *18, *19, *20, *21, *22, *24, *25, *26, *27, *28, *29, 2, 3, 4, 5, 6, 7, #	testpoint	testpoint	27								